Term 3 Practice Activity: C2 Uniform Random Variables

Contents

Evaluated criteria

- C2.

1 Introductory Uniform Distribution

Problem. Let $X \sim \text{Uniform}(0, 10)$.

Questions/Tasks. Find:

(a) the density value $f(x)$ on the support,
(b) the support interval of X,
(c) $P(2 \leq X \leq 7)$,
(d) $P(X > 8)$,
(e) $P(X \leq 8)$.

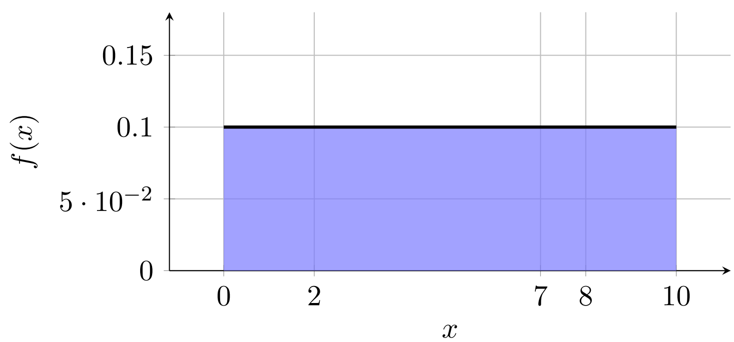
C2

(a) For $X \sim \text{Uniform}(0, 10)$,

$$f(x) = \frac{1}{10-0} = \frac{1}{10}, \qquad 0 \leq x \leq 10.$$

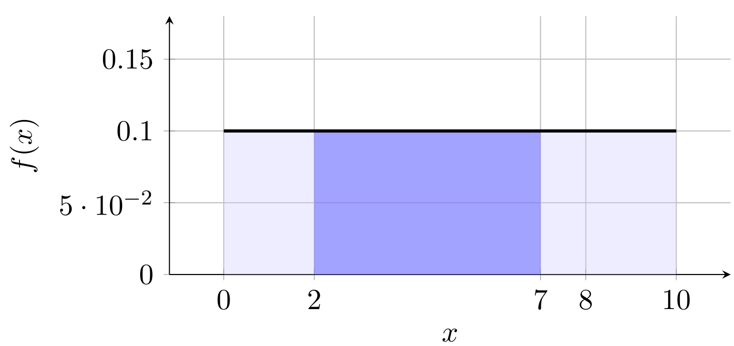
(b) The support is $[0, 10]$.

$$P(0 \leq X \leq 10) = 1.$$



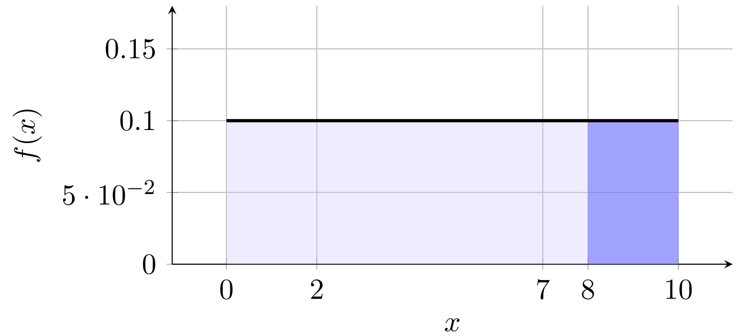
(c)

$$P(2 \leq X \leq 7) = \frac{7-2}{10} = \frac{1}{2}.$$



(d)

$$P(X > 8) = \frac{10-8}{10} = \frac{1}{5}.$$



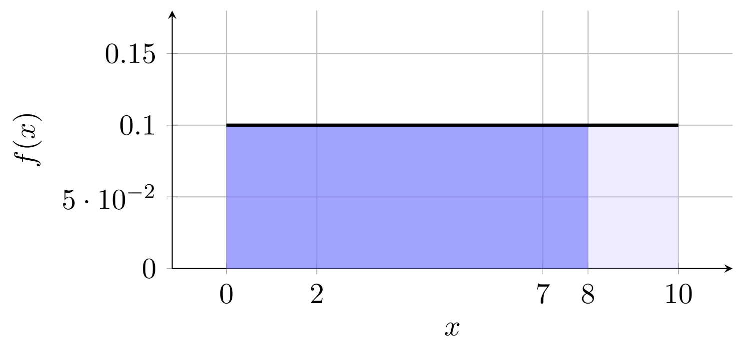
(e) For $P(X \leq 8)$, the standard approach gives

$$P(X \leq 8) = \frac{8-0}{10} = \frac{4}{5}.$$

Using the complement approach,

$$P(X \leq 8) = 1 - P(X > 8) = 1 - \frac{1}{5} = \frac{4}{5}.$$

So both approaches give the same value.



Interpretation. For a simple uniform model, probability is interval length divided by total length. Complements give the same result because the region $X \leq 8$ and the region $X > 8$ together make the whole support.

2 Introductory Uniform Distribution with Variable Endpoint

Problem. Let $X \sim \text{Uniform}(0, b)$, where $b > 8$.

Questions/Tasks. Find:

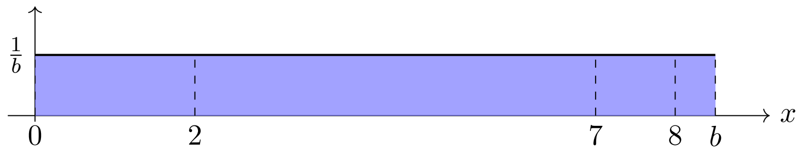
(a) the density value $f(x)$ on the support,
(b) the support interval of X,
(c) $P(2 \leq X \leq 7)$,
(d) $P(X > 8)$,
(e) $P(X \leq 8)$.

C2

(a) For $X \sim \text{Uniform}(0, b)$,

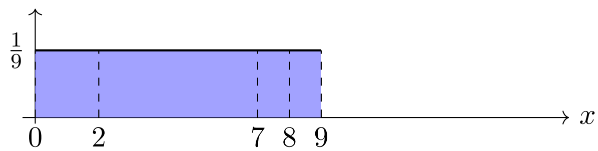
$$f(x) = \frac{1}{b-0} = \frac{1}{b}, \qquad 0 \leq x \leq b.$$

(b) The support is $[0, b]$.

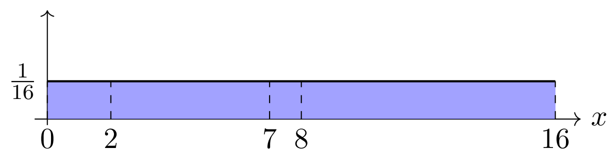


Consider the following cases obtained by assigning specific values to b.

Example with $b = 9$

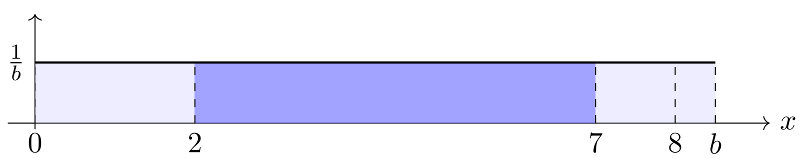


Example with $b = 16$



(c)

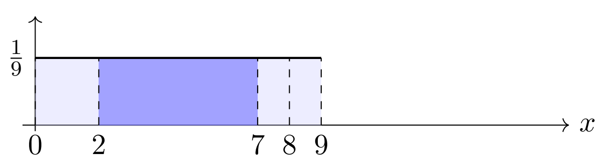
$$P(2 \leq X \leq 7) = \frac{7-2}{b} = \frac{5}{b}.$$



Consider the following cases obtained by assigning specific values to b.

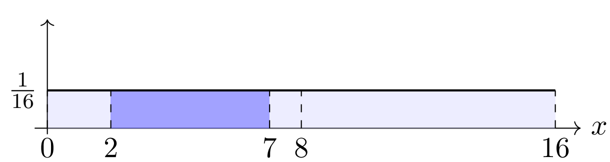
Example with $b = 9$

$$P(2 \leq X \leq 7) = \frac{5}{9}$$



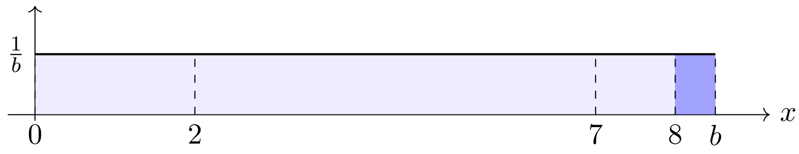
Example with $b = 16$

$$P(2 \leq X \leq 7) = \frac{5}{16}$$



(d)

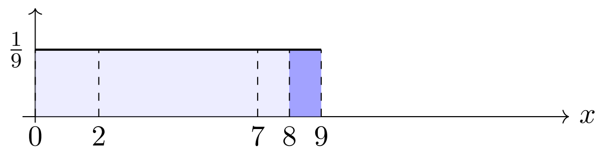
$$P(X > 8) = \frac{b-8}{b}.$$



Consider the following cases obtained by assigning specific values to b.

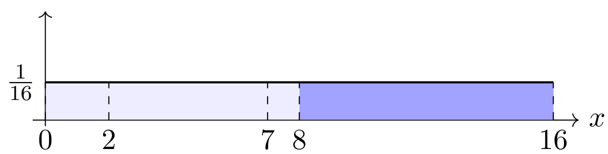
Example with $b = 9$

$$P(X > 8) = \frac{1}{9}$$



Example with $b = 16$

$$P(X > 8) = \frac{8}{16} = \frac{1}{2}$$



(e) The standard approach gives

$$P(X \leq 8) = \frac{8 - 0}{b} = \frac{8}{b}.$$

Using the complement approach,

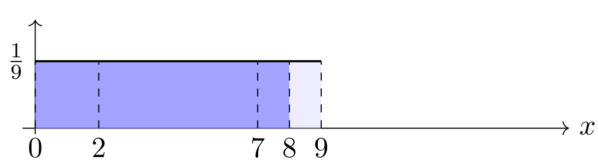
$$P(X \leq 8) = 1 - P(X > 8) = 1 - \frac{b - 8}{b} = \frac{b - (b - 8)}{b} = \frac{8}{b}.$$

So both approaches give the same value.

Consider the following cases obtained by assigning specific values to b.

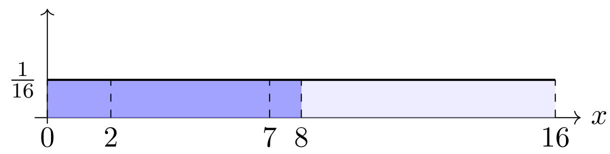
Example with $b = 9$

$$P(X \leq 8) = \frac{8}{9}$$



Example with $b = 16$

$$P(X \le 8) = \frac{8}{16} = \frac{1}{2}$$



Interpretation. For a simple uniform model with variable endpoint, probability is interval length divided by total length. Complements give the same result because the region $X \le 8$ and the region $X > 8$ together make the whole support.

3 Introductory Uniform Distribution with Variable Startpoint

Problem. Let $X \sim \text{Uniform}(a, 10)$, where $a < 2$.

Questions/Tasks. Find:

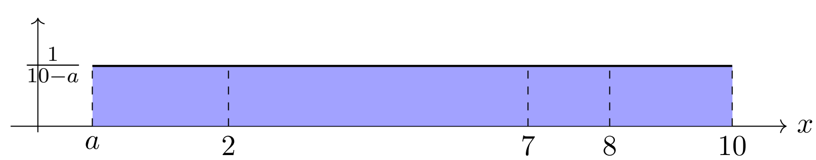
(a) the density value $f(x)$ on the support,
(b) the support interval of X,
(c) $P(2 \le X \le 7)$,
(d) $P(X > 8)$,
(e) $P(X \le 8)$.

C2

(a) For $X \sim \text{Uniform}(a, 10)$,

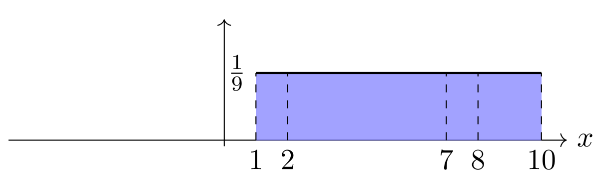
$$f(x) = \frac{1}{10 - a}, \qquad a \le x \le 10.$$

(b) The support is $[a, 10]$.

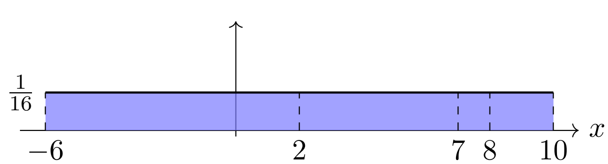


Consider the following cases obtained by assigning specific values to a.

Example with $a = 1$

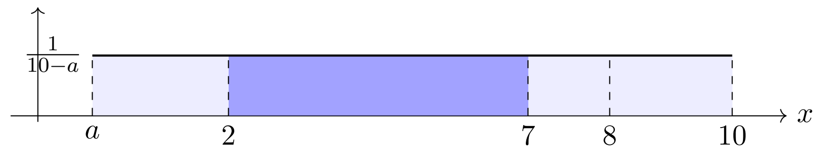


Example with $a = -6$



(c)

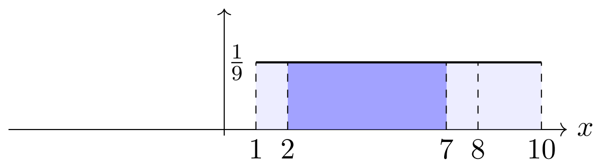
$$P(2 \leq X \leq 7) = \frac{7-2}{10-a} = \frac{5}{10-a}.$$



Consider the following cases obtained by assigning specific values to a.

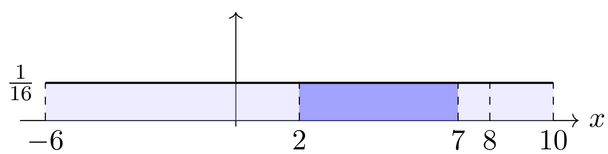
Example with $a = 1$

$$P(2 \leq X \leq 7) = \frac{5}{9}$$



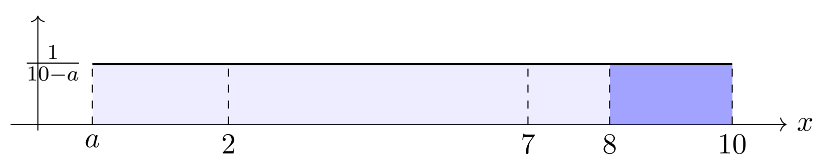
Example with $a = -6$

$$P(2 \leq X \leq 7) = \frac{5}{16}$$



(d)

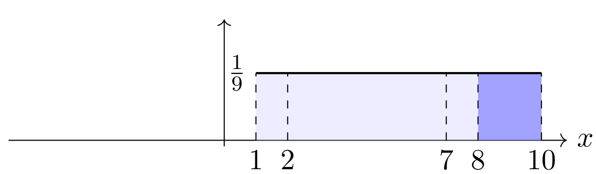
$$P(X > 8) = \frac{10-8}{10-a} = \frac{2}{10-a}.$$



Consider the following cases obtained by assigning specific values to a.

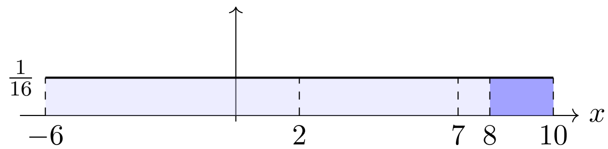
Example with $a = 1$

$$P(X > 8) = \frac{2}{9}$$



Example with $a = -6$

$$P(X > 8) = \frac{2}{16} = \frac{1}{8}$$



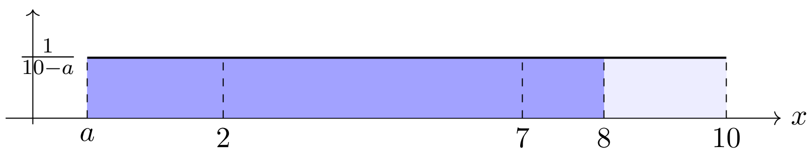
(e) The standard approach gives

$$P(X \le 8) = \frac{8-a}{10-a}.$$

Using the complement approach,

$$P(X \le 8) = 1 - P(X > 8) = 1 - \frac{2}{10-a} = \frac{10-a-2}{10-a} = \frac{8-a}{10-a}.$$

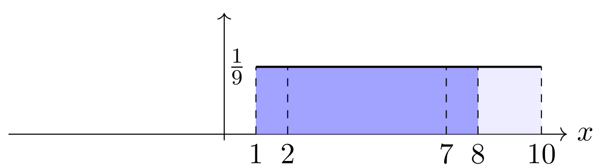
So both approaches give the same value.



Consider the following cases obtained by assigning specific values to a.

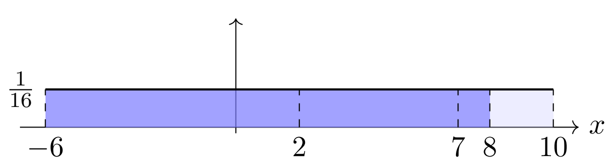
Example with $a = 1$

$$P(X \le 8) = \frac{7}{9}$$



Example with $a = -6$

$$P(X \le 8) = \frac{14}{16} = \frac{7}{8}$$



Interpretation. For a simple uniform model with variable startpoint, probability is interval length divided by total length. Complements give the same result because the region $X \le 8$ and the region $X > 8$ together make the whole support.

4 Introductory Uniform Distribution with Variable Endpoints

Problem. Let $X \sim \text{Uniform}(a, b)$, where $a < 2 < 7 < 8 < b$.

Questions/Tasks. Find:

(a) the density value $f(x)$ on the support,

(b) the support interval of X,

(c) $P(2 \leq X \leq 7)$,

(d) $P(X > 8)$,

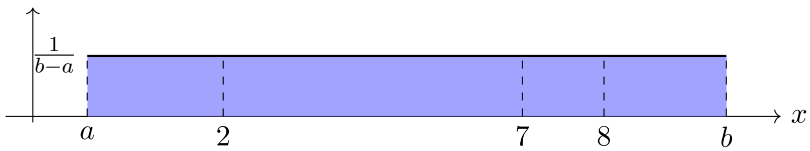
(e) $P(X \leq 8)$.

C2

(a) For $X \sim \text{Uniform}(a, b)$,

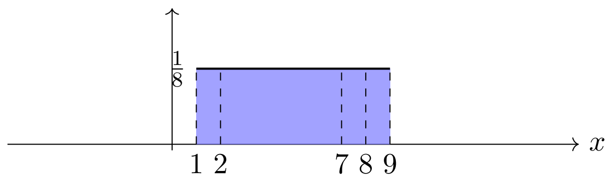
$$f(x) = \frac{1}{b-a}, \qquad a \leq x \leq b.$$

(b) The support is $[a, b]$.

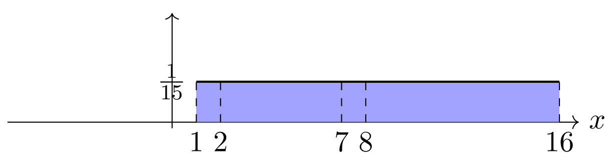


Consider the following cases obtained by assigning specific values to the endpoints.

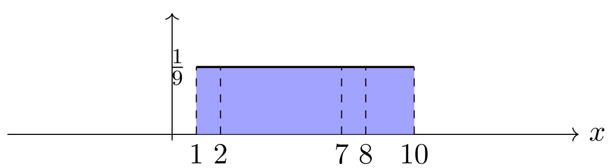
Startpoint $a = 1$, *endpoint* $b = 9$



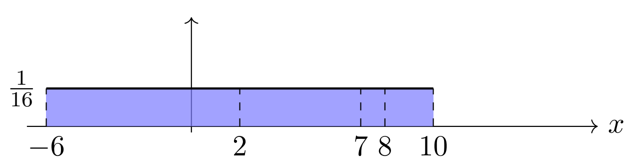
Startpoint $a = 1$, *endpoint* $b = 16$



Endpoint $b = 10$, *startpoint* $a = 1$

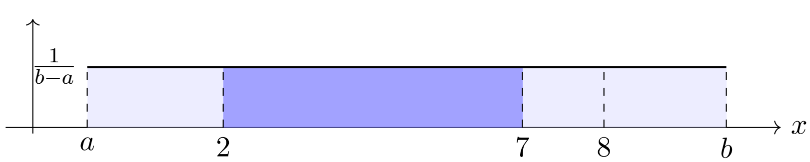


Endpoint $b = 10$, *startpoint* $a = -6$



(c)

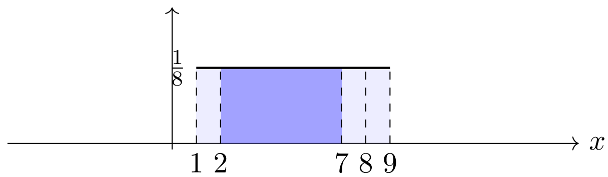
$$P(2 \leq X \leq 7) = \frac{7-2}{b-a} = \frac{5}{b-a}.$$



Consider the following cases for part (c).

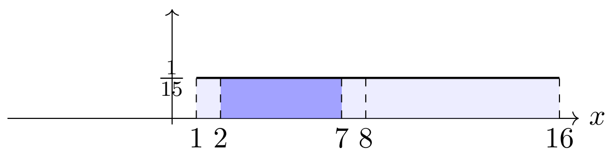
$$\textit{Startpoint } a = 1, \ \textit{endpoint } b = 9$$

$$P(2 \le X \le 7) = \frac{5}{8}$$



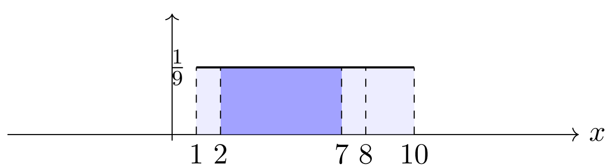
$$\textit{Startpoint } a = 1, \ \textit{endpoint } b = 16$$

$$P(2 \le X \le 7) = \frac{5}{15} = \frac{1}{3}$$



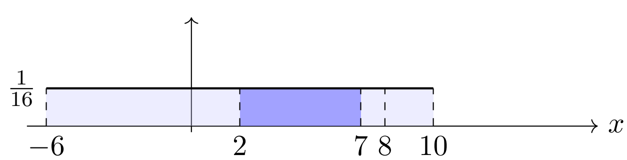
$$\textit{Endpoint } b = 10, \ \textit{startpoint } a = 1$$

$$P(2 \le X \le 7) = \frac{5}{9}$$



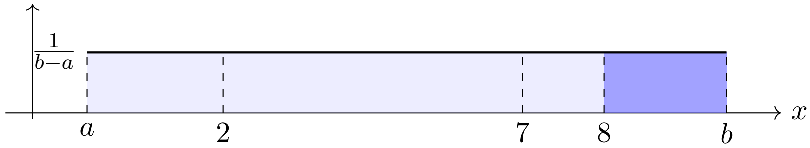
$$\textit{Endpoint } b = 10, \ \textit{startpoint } a = -6$$

$$P(2 \le X \le 7) = \frac{5}{16}$$



(d)

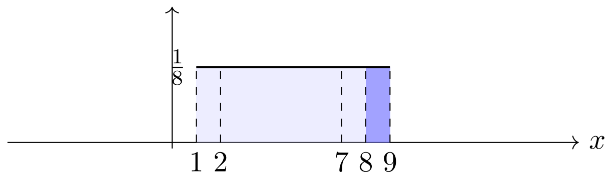
$$P(X > 8) = \frac{b-8}{b-a}.$$



Consider the following cases for part (d).

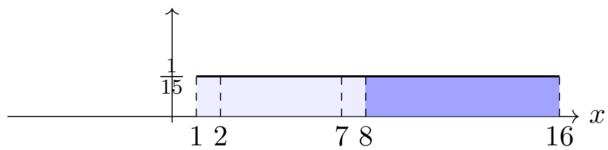
$$\textit{Startpoint } a = 1, \ \textit{endpoint } b = 9$$

$$P(X > 8) = \frac{1}{8}$$



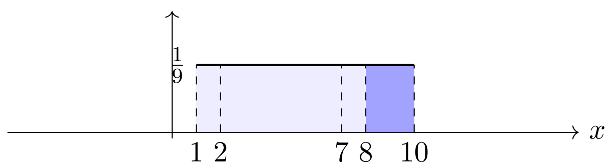
$$\textit{Startpoint } a = 1, \ \textit{endpoint } b = 16$$

$$P(X > 8) = \frac{8}{15}$$



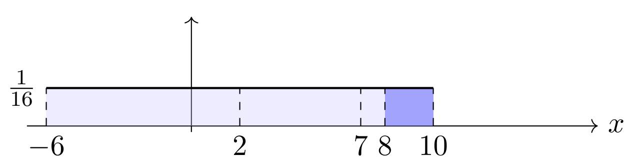
$$\textit{Endpoint } b = 10, \ \textit{startpoint } a = 1$$

$$P(X > 8) = \frac{2}{9}$$



$$\textit{Endpoint } b = 10, \ \textit{startpoint } a = -6$$

$$P(X > 8) = \frac{2}{16} = \frac{1}{8}$$



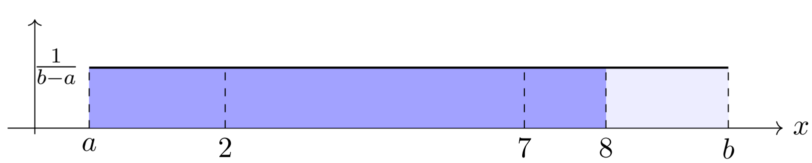
(e) The standard approach gives

$$P(X \leq 8) = \frac{8-a}{b-a}.$$

Using the complement approach,

$$P(X \leq 8) = 1 - P(X > 8) = 1 - \frac{b-8}{b-a} = \frac{b-a-(b-8)}{b-a} = \frac{8-a}{b-a}.$$

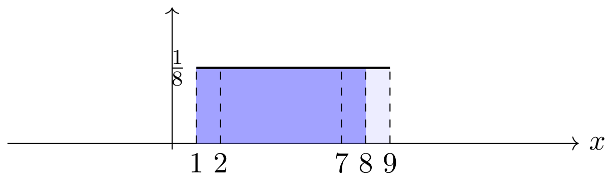
So both approaches give the same value.



Consider the following cases for part (e).

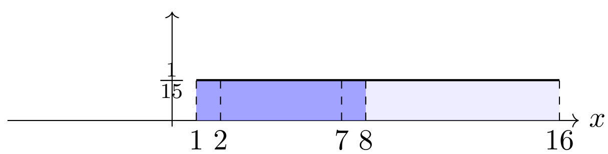
$$\textit{Startpoint } a = 1, \ \textit{endpoint } b = 9$$

$$P(X \leq 8) = \frac{7}{8}$$



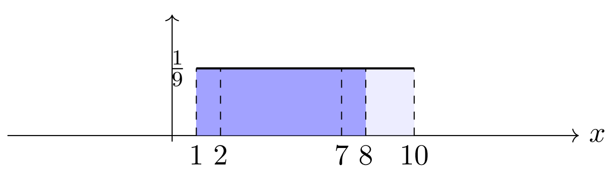
$$\textit{Startpoint } a = 1, \ \textit{endpoint } b = 16$$

$$P(X \leq 8) = \frac{7}{15}$$



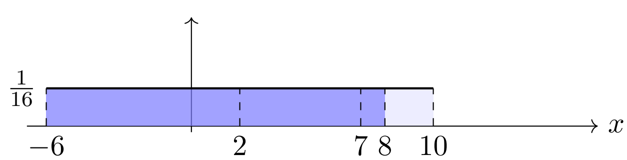
$$\textit{Endpoint } b = 10, \ \textit{startpoint } a = 1$$

$$P(X \leq 8) = \frac{7}{9}$$



$$\textit{Endpoint } b = 10, \ \textit{startpoint } a = -6$$

$$P(X \leq 8) = \frac{14}{16} = \frac{7}{8}$$



Interpretation. For a simple uniform model with variable endpoints, probability is interval length divided by total length. Complements give the same result because the region $X \leq 8$ and the region $X > 8$ together make the whole support.

5 Two-Interval Piecewise Uniform (Basic)

Problem. Let X have density

$$f(x) = \begin{cases} \frac{1}{5}, & 0 \le x < 4, \\ \frac{1}{30}, & 4 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

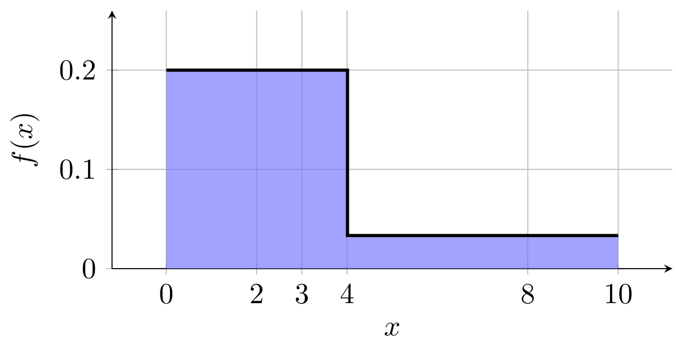
Questions/Tasks. Find:

(a) $P(X < 3)$,
(b) $P(X \ge 3)$,
(c) $P(2 \le X \le 8)$.

C2

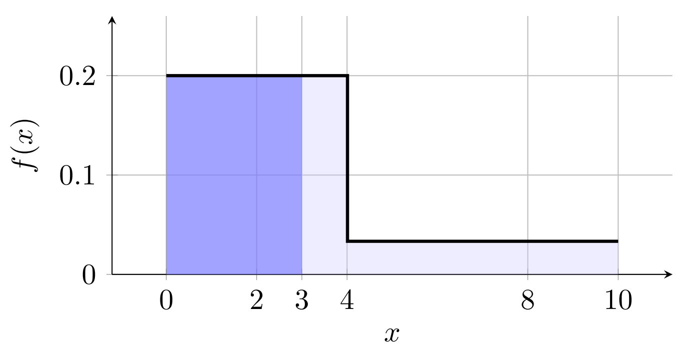
Function definition. Check total area:

$$4\left(\frac{1}{5}\right) + 6\left(\frac{1}{30}\right) = \frac{4}{5} + \frac{1}{5} = 1.$$



(a)

$$P(X < 3) = 3\left(\frac{1}{5}\right) = \frac{3}{5}.$$



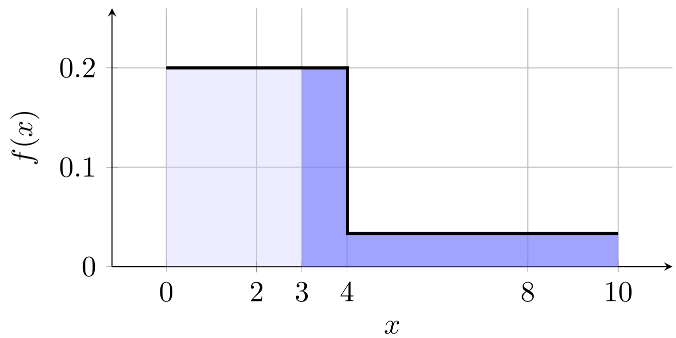
(b) Using the standard approach,

$$P(X \ge 3) = 1\left(\frac{1}{5}\right) + 6\left(\frac{1}{30}\right) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}.$$

Using the complement approach,

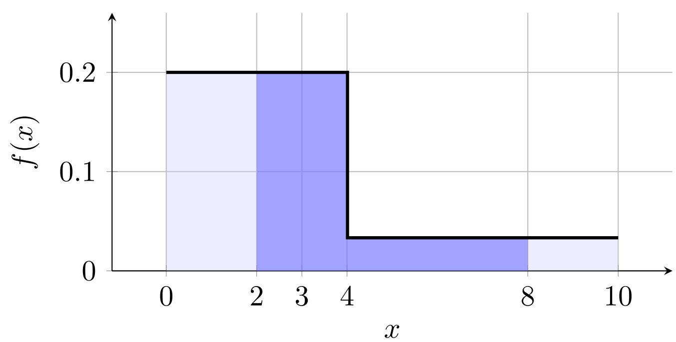
$$P(X \geq 3) = 1 - P(X < 3) = 1 - \frac{3}{5} = \frac{2}{5}.$$

So both approaches give the same value.



(c)

$$P(2 \leq X \leq 8) = 2\left(\frac{1}{5}\right) + 4\left(\frac{1}{30}\right) = \frac{2}{5} + \frac{2}{15} = \frac{8}{15}.$$



Interpretation. Most probability mass is on $[0, 4)$ because its density height is much larger.

6 Two-Interval Piecewise Uniform (Basic)

Problem. Let X have density

$$f(x) = \begin{cases} \frac{1}{6}, & -2 \leq x < 1, \\ \frac{1}{10}, & 1 \leq x \leq 6, \\ 0, & \text{otherwise.} \end{cases}$$

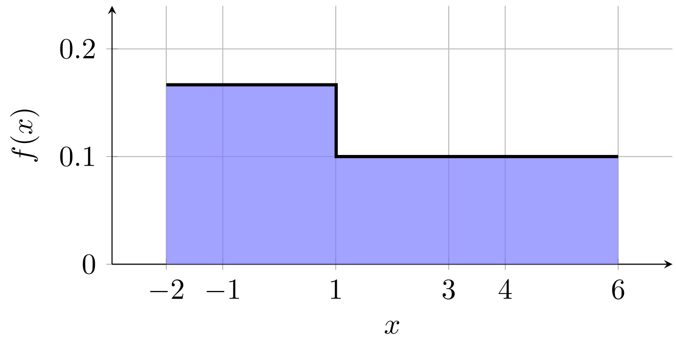
Questions/Tasks. Find:

(a) $P(-1 \leq X \leq 4)$,
(b) $P(X > 3)$,
(c) $P(X \leq 3)$.

C2

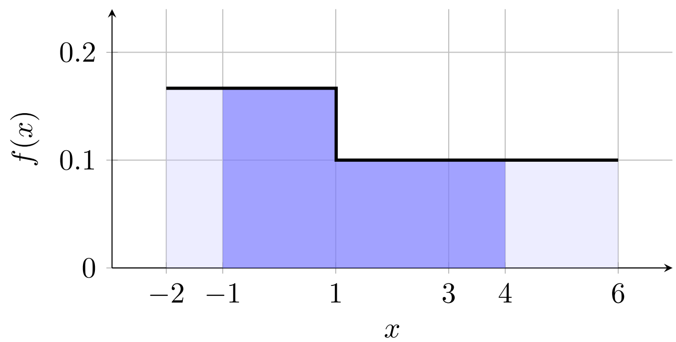
Check total area:

$$3\left(\frac{1}{6}\right) + 5\left(\frac{1}{10}\right) = \frac{1}{2} + \frac{1}{2} = 1.$$



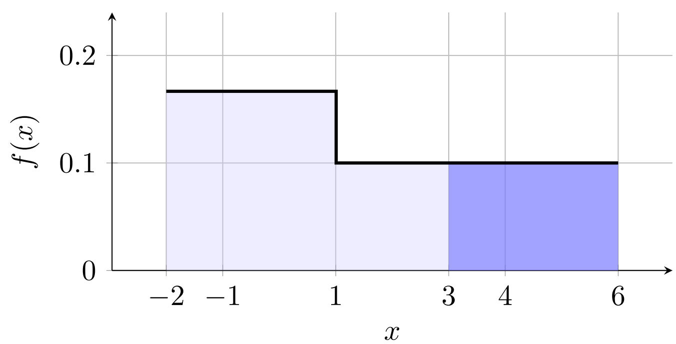
(a)

$$P(-1 \le X \le 4) = 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{10}\right) = \frac{1}{3} + \frac{3}{10} = \frac{19}{30}.$$



(b)

$$P(X > 3) = 3\left(\frac{1}{10}\right) = \frac{3}{10}.$$



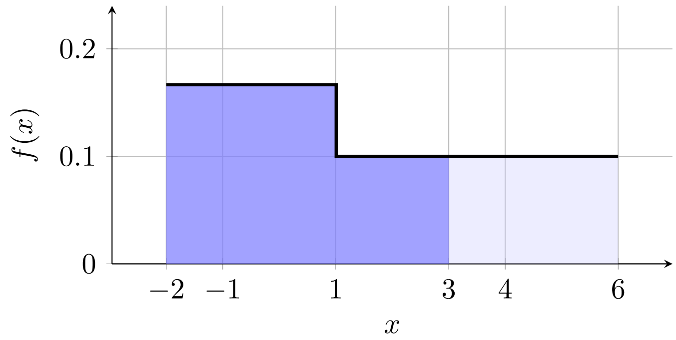
(c) Using the standard approach,

$$P(X \le 3) = 3\left(\frac{1}{6}\right) + 2\left(\frac{1}{10}\right) = \frac{1}{2} + \frac{1}{5} = \frac{7}{10}.$$

Using the complement approach,

$$P(X \le 3) = 1 - P(X > 3) = 1 - \frac{3}{10} = \frac{7}{10}.$$

So both approaches give the same value.



Interpretation. To find each probability, add the rectangular areas over the relevant subintervals. In part (c), the complement method works because the regions $X \le 3$ and $X > 3$ together make the whole support.

7 Two-Interval Piecewise Uniform with Unique Variable Endpoint

Problem. Let X have density

$$f(x) = \begin{cases} \dfrac{1}{12}, & -2 \le x < 1, \\ \dfrac{1}{8}, & 1 \le x \le b, \\ 0, & \text{otherwise,} \end{cases}$$

where $b > 1$.

Questions/Tasks. Find:

(a) $P(X < -1)$,
(b) $P(X > 3)$,
(c) $P(-1 \le X \le 3)$.

C2

For a valid density, the total area must be 1:

$$3\left(\frac{1}{12}\right) + (b-1)\left(\frac{1}{8}\right) = 1.$$

So

$$\frac{1}{4} + \frac{b-1}{8} = 1.$$

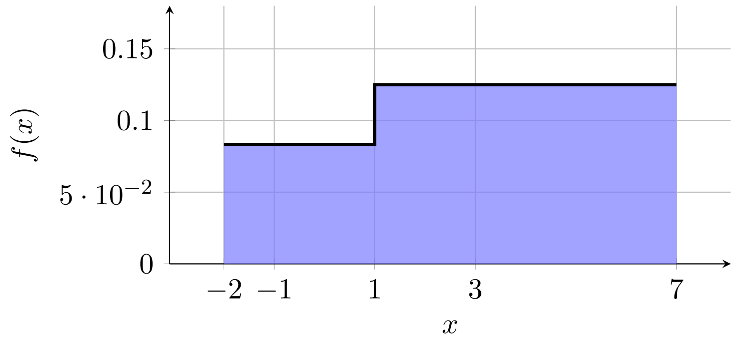
Thus

$$\frac{b-1}{8} = \frac{3}{4},$$

and therefore

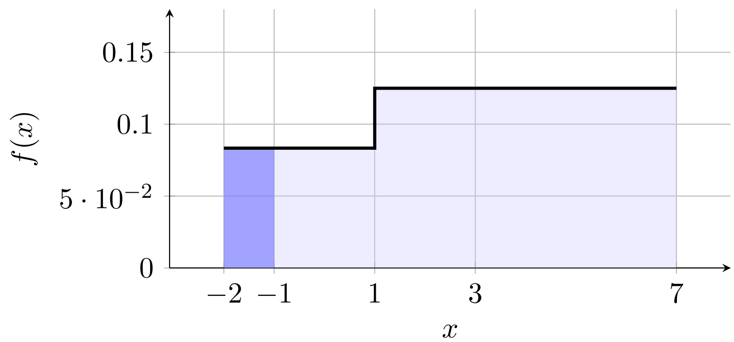
$$b - 1 = 6, \qquad b = 7.$$

Now use $b = 7$.



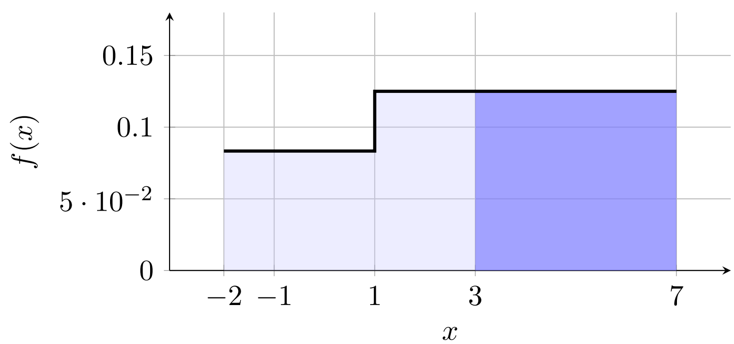
(a)

$$P(X < -1) = 1\left(\frac{1}{12}\right) = \frac{1}{12}.$$



(b)

$$P(X > 3) = 4\left(\frac{1}{8}\right) = \frac{1}{2}.$$



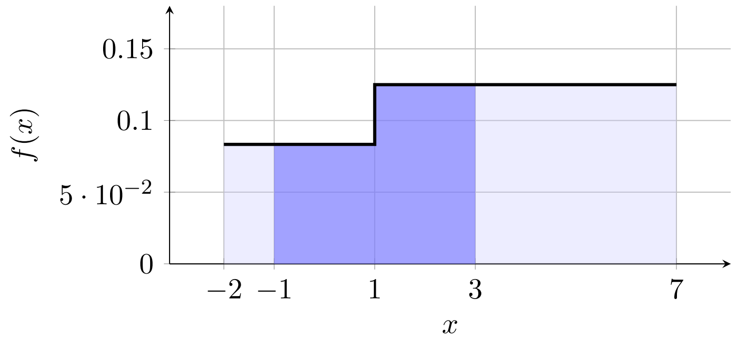
(c) Using the standard approach,

$$P(-1 \leq X \leq 3) = 2\left(\frac{1}{12}\right) + 2\left(\frac{1}{8}\right) = \frac{1}{6} + \frac{1}{4} = \frac{5}{12}.$$

Using the complement approach,

$$P(-1 \leq X \leq 3) = 1 - P(X < -1) - P(X > 3) = 1 - \frac{1}{12} - \frac{1}{2} = \frac{5}{12}.$$

So both approaches give the same value.



8 Two-Interval Piecewise Uniform with Unique Variable Startpoint

Problem. Let X have density

$$f(x) = \begin{cases} \frac{1}{6}, & a \le x < 1, \\ \frac{1}{10}, & 1 \le x \le 6, \\ 0, & \text{otherwise}, \end{cases}$$

where $a < 1$.

Questions/Tasks. Find:

(a) the value of a that makes this a valid density,
(b) $P(-1 \le X \le 4)$,
(c) $P(X > 3)$,
(d) $P(X < 1)$.

C2

For a valid density, the total area must be 1:

$$(1-a)\left(\frac{1}{6}\right) + 5\left(\frac{1}{10}\right) = 1.$$

So

$$\frac{1-a}{6} + \frac{1}{2} = 1.$$

Thus

$$\frac{1-a}{6} = \frac{1}{2},$$

and therefore

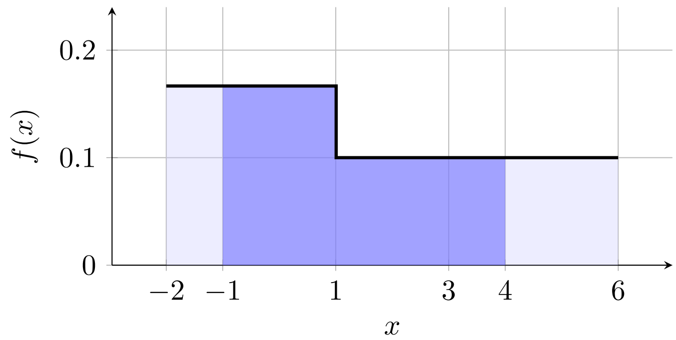
$$1 - a = 3, \qquad a = -2.$$

Now use $a = -2$.

$$P(-1 \le X \le 4) = 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{10}\right) = \frac{1}{3} + \frac{3}{10} = \frac{19}{30}.$$

$$P(X > 3) = 3\left(\frac{1}{10}\right) = \frac{3}{10}.$$

$$P(X < 1) = 3\left(\frac{1}{6}\right) = \frac{1}{2}.$$



Interpretation. Because the heights are fixed, there is only one startpoint that makes the total area equal to 1. After finding that startpoint, probabilities are found by adding the relevant rectangular areas.

9 Two-Interval Piecewise Uniform with Variable Endpoint

Problem. Let X have density

$$f(x) = \begin{cases} \frac{1}{6}, & -2 \le x < 1, \\ \frac{1}{2(b-1)}, & 1 \le x \le b, \\ 0, & \text{otherwise}, \end{cases} \qquad \text{where } b > 4.$$

Questions/Tasks. Find:

(a) $P(-1 \le X \le 4)$,
(b) $P(X > 3)$,
(c) $P(X < 1)$.

C2

Check total area:

$$3\left(\frac{1}{6}\right) + (b-1)\left(\frac{1}{2(b-1)}\right) = \frac{1}{2} + \frac{1}{2} = 1.$$

So this is a valid density.

For part (a), split the interval at $x = 1$:

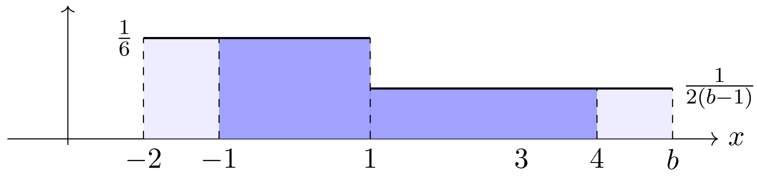
$$P(-1 \le X \le 4) = 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{2(b-1)}\right) = \frac{1}{3} + \frac{3}{2(b-1)}.$$

For part (b),

$$P(X > 3) = (b-3)\left(\frac{1}{2(b-1)}\right) = \frac{b-3}{2(b-1)}.$$

For part (c),

$$P(X < 1) = 3\left(\frac{1}{6}\right) = \frac{1}{2}.$$



Consider the following cases obtained by assigning specific values to b.

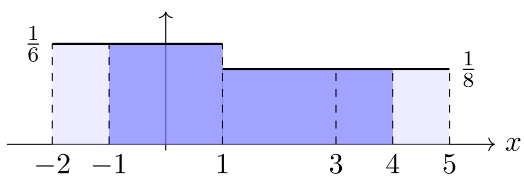
Example with $b = 5$

$$\frac{1}{2(b-1)} = \frac{1}{2(5-1)} = \frac{1}{8}$$

(a) $P(-1 \le X \le 4) = \frac{1}{3} + \frac{3}{2(5-1)} = \frac{1}{3} + \frac{3}{8} = \frac{17}{24}$

(b) $P(X > 3) = \frac{5-3}{2(5-1)} = \frac{1}{4}$

(c) $P(X < 1) = \frac{1}{2}$



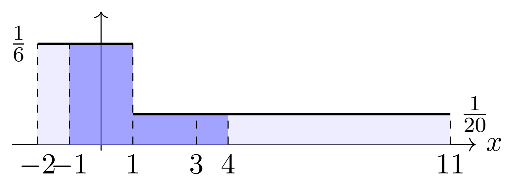
Example with $b = 11$

$$\frac{1}{2(b-1)} = \frac{1}{2(11-1)} = \frac{1}{20}$$

(a) $P(-1 \le X \le 4) = \frac{1}{3} + \frac{3}{2(11-1)} = \frac{1}{3} + \frac{3}{20} = \frac{29}{60}$

(b) $P(X > 3) = \frac{11-3}{2(11-1)} = \frac{8}{20} = \frac{2}{5}$

(c) $P(X < 1) = \frac{1}{2}$



Interpretation. To find each probability, add the rectangular areas over the relevant subintervals.

10 Two-Interval Piecewise Uniform with Variable Start

Problem. Let X have density

$$f(x) = \begin{cases} \dfrac{1}{2(1-a)}, & a \le x < 1, \\ \dfrac{1}{10}, & 1 \le x \le 6, \\ 0, & \text{otherwise}, \end{cases} \qquad \text{where } a < -1.$$

Questions/Tasks. Find:

(a) $P(-1 \le X \le 4)$,
(b) $P(X > 3)$,
(c) $P(X < 1)$.

C2

Check total area:

$$(1-a)\left(\frac{1}{2(1-a)}\right) + 5\left(\frac{1}{10}\right) = \frac{1}{2} + \frac{1}{2} = 1.$$

So this is a valid density.

For part (a), split the interval at $x = 1$:

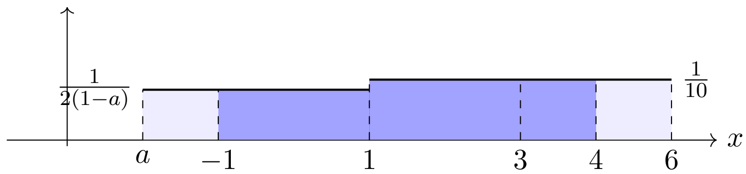
$$P(-1 \le X \le 4) = 2\left(\frac{1}{2(1-a)}\right) + 3\left(\frac{1}{10}\right) = \frac{1}{1-a} + \frac{3}{10}.$$

For part (b),

$$P(X > 3) = 3\left(\frac{1}{10}\right) = \frac{3}{10}.$$

For part (c),

$$P(X < 1) = (1-a)\left(\frac{1}{2(1-a)}\right) = \frac{1}{2}.$$



Consider the following cases obtained by assigning specific values to a.

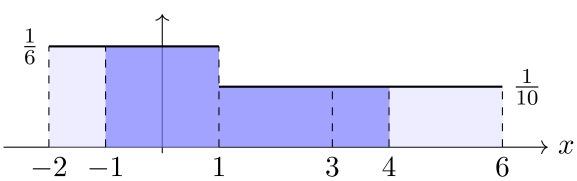
Example with $a = -2$

$$\frac{1}{2(1-a)} = \frac{1}{2(1-(-2))} = \frac{1}{6}$$

(a) $P(-1 \le X \le 4) = \dfrac{1}{1-(-2)} + \dfrac{3}{10} = \dfrac{1}{3} + \dfrac{3}{10} = \dfrac{19}{30}$

(b) $P(X > 3) = \dfrac{3}{10}$

(c) $P(X < 1) = \dfrac{1}{2}$



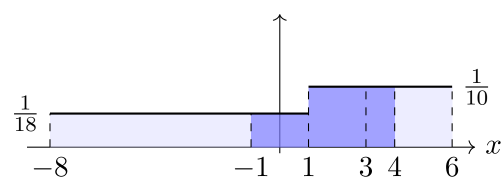
Example with $a = -8$

$$\frac{1}{2(1-a)} = \frac{1}{2(1-(-8))} = \frac{1}{18}$$

(a) $P(-1 \le X \le 4) = \dfrac{1}{1-(-8)} + \dfrac{3}{10} = \dfrac{1}{9} + \dfrac{3}{10} = \dfrac{37}{90}$

(b) $P(X > 3) = \dfrac{3}{10}$

(c) $P(X < 1) = \dfrac{1}{2}$



Interpretation. To find each probability, add the rectangular areas over the relevant subintervals.

11 Three-Interval Piecewise Uniform (Combining Regions)

Problem. Let X have density

$$f(x) = \begin{cases} \frac{1}{10}, & 0 \le x < 2, \\ \frac{1}{5}, & 2 \le x < 5, \\ \frac{1}{10}, & 5 \le x \le 7, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks. Find:

(a) $P(X < 2 \text{ or } X > 5)$,
(b) $P(2 \le X \le 4)$,
(c) Which is more likely: $X \in [0, 2)$ or $X \in [2, 5)$? Explain.

C2

Check total area:

$$2\left(\frac{1}{10}\right) + 3\left(\frac{1}{5}\right) + 2\left(\frac{1}{10}\right) = \frac{1}{5} + \frac{3}{5} + \frac{1}{5} = 1.$$

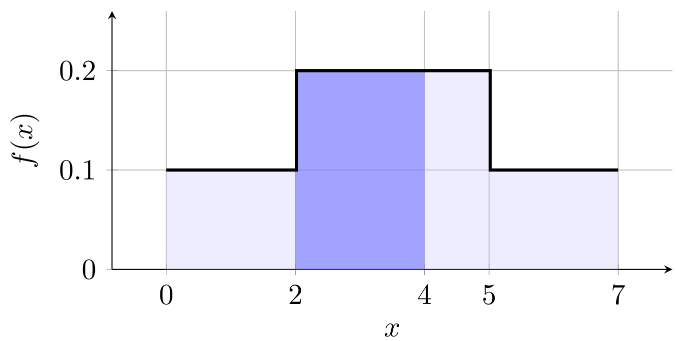
$$P(X < 2 \text{ or } X > 5) = 2\left(\frac{1}{10}\right) + 2\left(\frac{1}{10}\right) = \frac{2}{5}.$$

$$P(2 \le X \le 4) = 2\left(\frac{1}{5}\right) = \frac{2}{5}.$$

Also,

$$P(0 \le X < 2) = \frac{1}{5}, \qquad P(2 \le X < 5) = \frac{3}{5}.$$

So $[2, 5)$ is more likely.



Interpretation. Even with equal widths, a taller density rectangle gives larger probability.

12 Piecewise Uniform with Complement Reasoning

Problem. Let X have density

$$f(x) = \begin{cases} \frac{3}{20}, & 10 \le x < 14, \\ \frac{1}{10}, & 14 \le x < 18, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks. Find:

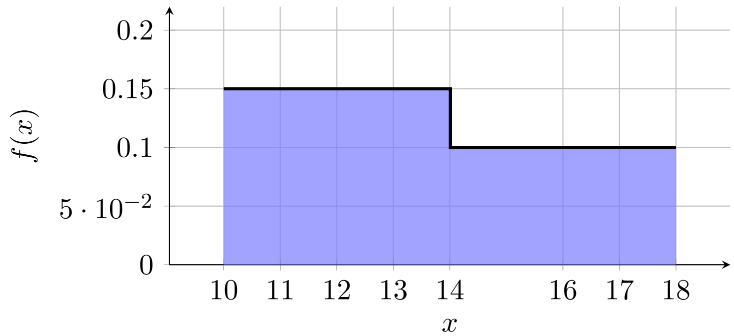
(a) $P(11 \le X \le 13 \text{ or } 16 \le X \le 18)$,
(b) $P(X < 12 \text{ or } X > 17)$,
(c) $P(12 \le X \le 17)$ using a complement.

C2

Check total area:

$$4\left(\frac{3}{20}\right) + 4\left(\frac{1}{10}\right) = \frac{3}{5} + \frac{2}{5} = 1.$$

So this is a valid density function.



a)

The two intervals are disjoint, so add their areas.

From 11 to 13, the width is 2 and the height is $\frac{3}{20}$, so

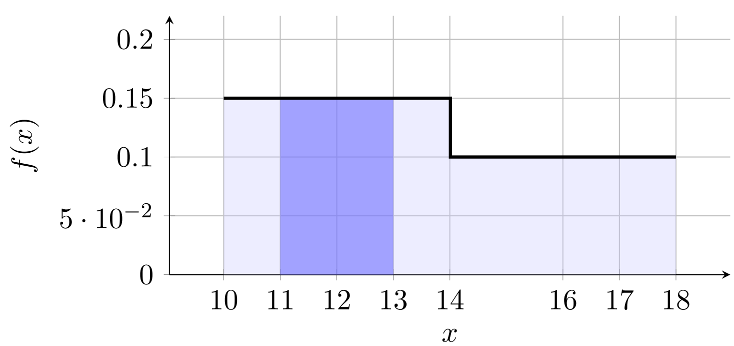
$$P(11 \leq X \leq 13) = 2\left(\frac{3}{20}\right) = \frac{3}{10}.$$

From 16 to 18, the width is 2 and the height is $\frac{1}{10}$, so

$$P(16 \leq X \leq 18) = 2\left(\frac{1}{10}\right) = \frac{1}{5}.$$

Therefore,

$$P(11 \leq X \leq 13 \text{ or } 16 \leq X \leq 18) = \frac{3}{10} + \frac{1}{5} = \frac{1}{2}.$$



b)

Again, the two regions are disjoint.

For $X < 12$, the interval is from 10 to 12, with width 2 and height $\frac{3}{20}$:

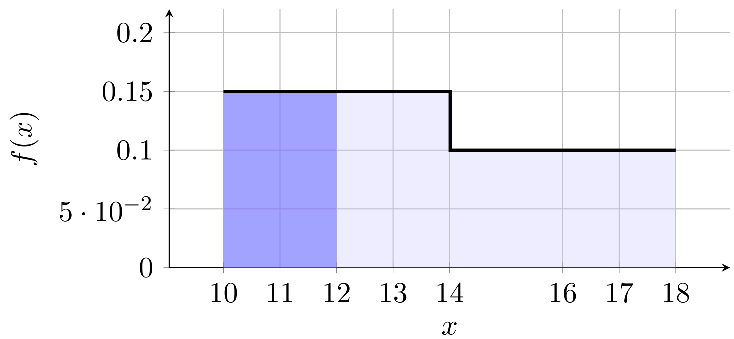
$$P(X < 12) = 2\left(\frac{3}{20}\right) = \frac{3}{10}.$$

For $X > 17$, the interval is from 17 to 18, with width 1 and height $\frac{1}{10}$:

$$P(X > 17) = 1\left(\frac{1}{10}\right) = \frac{1}{10}.$$

So

$$P(X < 12 \text{ or } X > 17) = \frac{3}{10} + \frac{1}{10} = \frac{2}{5}.$$



c)

Use the complement of the event from part (b). The interval $12 \le X \le 17$ is exactly the remaining part of the support after removing $X < 12$ and $X > 17$.

Therefore,

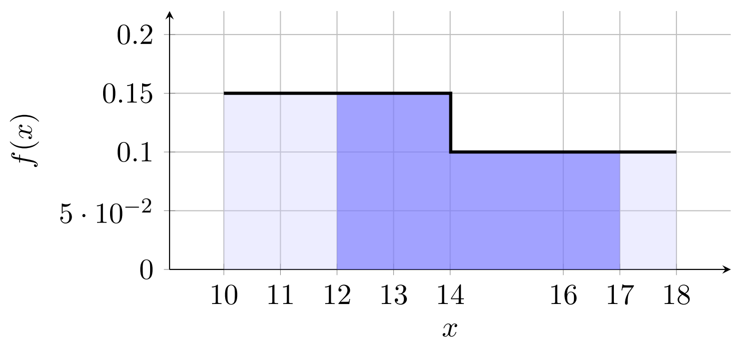
$$P(12 \le X \le 17) = 1 - P(X < 12 \text{ or } X > 17).$$

From part (b),

$$P(X < 12 \text{ or } X > 17) = \frac{2}{5}.$$

So

$$P(12 \le X \le 17) = 1 - \frac{2}{5} = \frac{3}{5}.$$



Interpretation. Complement reasoning avoids repeating long area sums.

13 Piecewise Uniform with Unknown k

Problem. Let X have density

$$f(x) = \begin{cases} 2k, & 0 \le x < 3, \\ k, & 3 \le x \le 8, \\ 0, & \text{otherwise}, \end{cases}$$

where $k > 0$.

Questions/Tasks. Find:

(a) the value of k so that f is a valid PDF,
(b) $P(X < 5)$,
(c) $P(6 \le X \le 8)$.

C2

a)

For validity, total area must be 1:

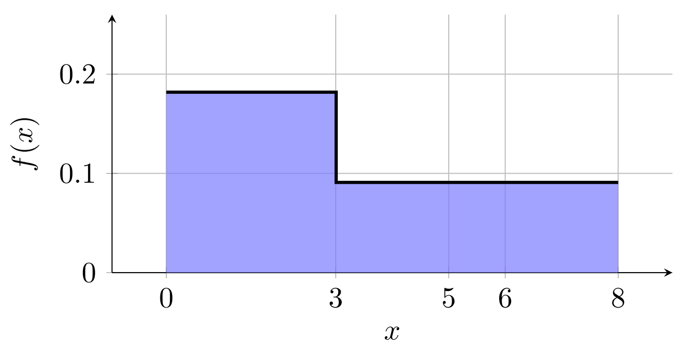
$$3(2k) + 5(k) = 1.$$

So

$$6k + 5k = 1 \qquad \Rightarrow \qquad 11k = 1 \qquad \Rightarrow \qquad k = \frac{1}{11}.$$

Therefore, the density becomes

$$f(x) = \begin{cases} \frac{2}{11}, & 0 \le x < 3, \\ \frac{1}{11}, & 3 \le x \le 8, \\ 0, & \text{otherwise}. \end{cases}$$



b)

To find $P(X < 5)$, it is easier to use the complement.

First compute the probability of the opposite event, $X \ge 5$. On the interval from 5 to 8, the density is constant at $\frac{1}{11}$. The width is

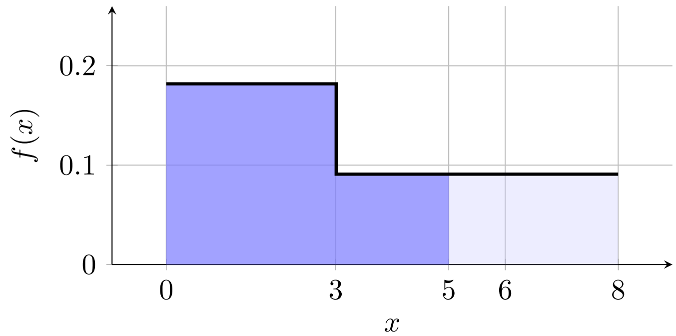
$$8 - 5 = 3,$$

so

$$P(X \geq 5) = 3\left(\frac{1}{11}\right) = \frac{3}{11}.$$

Therefore,

$$P(X < 5) = 1 - P(X \geq 5) = 1 - \frac{3}{11} = \frac{8}{11}.$$



c)

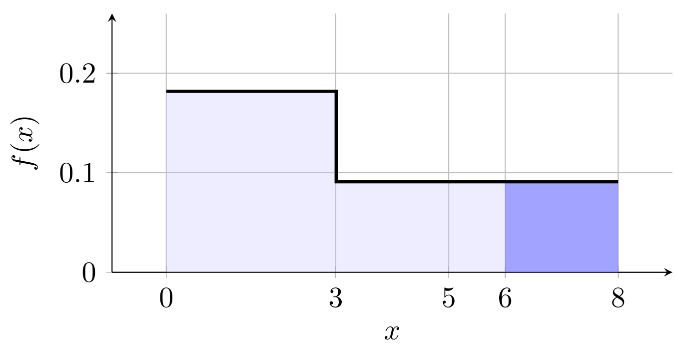
On the interval from 6 to 8, the density is constant at $\frac{1}{11}$.

The width is

$$8 - 6 = 2,$$

so

$$P(6 \leq X \leq 8) = 2\left(\frac{1}{11}\right) = \frac{2}{11}.$$



14 Piecewise Uniform with Unknown k and Validity Check

Problem. Let X have density

$$f(x) = \begin{cases} k, & 0 \leq x < 2, \\ 2k, & 2 \leq x < 5, \\ \frac{k}{2}, & 5 \leq x \leq 9, \\ 0, & \text{otherwise}, \end{cases}$$

where k is a constant.

Questions/Tasks. Find:

(a) k,
(b) verify that all density values are valid for a PDF,
(c) $P(1 \le X \le 6)$.

C2

Total area condition:

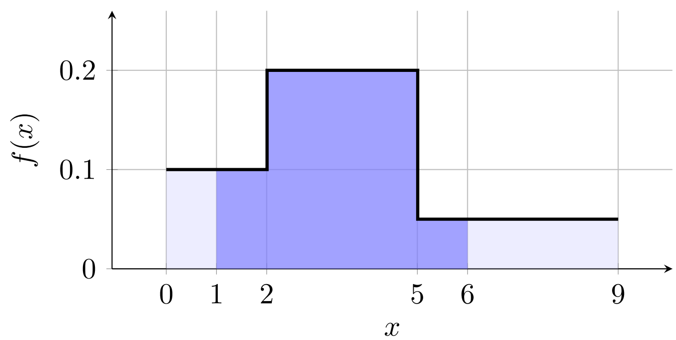
$$2(k) + 3(2k) + 4\left(\frac{k}{2}\right) = 1 \Rightarrow 2k + 6k + 2k = 1 \Rightarrow 10k = 1 \Rightarrow k = \frac{1}{10}.$$

Then heights are

$$\frac{1}{10}, \quad \frac{1}{5}, \quad \frac{1}{20},$$

all nonnegative, so valid.

$$P(1 \le X \le 6) = 1\left(\frac{1}{10}\right) + 3\left(\frac{1}{5}\right) + 1\left(\frac{1}{20}\right) = \frac{2}{20} + \frac{12}{20} + \frac{1}{20} = \frac{3}{4}.$$



Interpretation. A valid PDF needs both nonnegative heights and total area exactly 1.

15 Four-Interval Piecewise Uniform (Organizing Across Many Intervals)

Problem. Let X have density

$$f(x) = \begin{cases} \frac{3}{10}, & 0 \le x < 1, \\ \frac{1}{5}, & 1 \le x < 3, \\ \frac{1}{10}, & 3 \le x < 5, \\ \frac{1}{10}, & 5 \le x \le 6, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks. Find:

(a) $P(0.5 \le X \le 5.5)$,
(b) compare $P(X < 3)$ and $P(X \ge 3)$,
(c) identify where most of the probability mass is concentrated.

C2

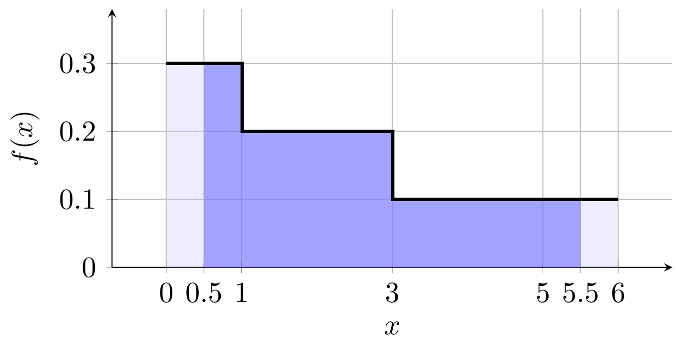
Check total area:

$$1\left(\frac{3}{10}\right)+2\left(\frac{1}{5}\right)+2\left(\frac{1}{10}\right)+1\left(\frac{1}{10}\right)=\frac{3}{10}+\frac{4}{10}+\frac{2}{10}+\frac{1}{10}=1.$$

$$P(0.5\le X\le 5.5)=0.5\left(\frac{3}{10}\right)+2\left(\frac{1}{5}\right)+2\left(\frac{1}{10}\right)+0.5\left(\frac{1}{10}\right)=\frac{3}{20}+\frac{8}{20}+\frac{4}{20}+\frac{1}{20}=\frac{4}{5}.$$

$$P(X<3)=\frac{3}{10}+\frac{4}{10}=\frac{7}{10},\qquad P(X\ge 3)=\frac{3}{10}.$$

Most mass is below 3.



Interpretation. Careful partitioning of a long interval keeps multi-step area sums accurate.

16 Structured Piecewise Uniform with Parameter k

Problem. Let X have density

$$f(x)=\begin{cases} k, & -3\le x<0,\\ 2k, & 0\le x<2,\\ \dfrac{1}{12}, & 2\le x\le 6,\\ 0, & \text{otherwise.}\end{cases}$$

Questions/Tasks. Find:

(a) k,

(b) $P(-1\le X\le 4)$,

(c) which is larger: $P(X<0)$ or $P(X\ge 2)$?

C2

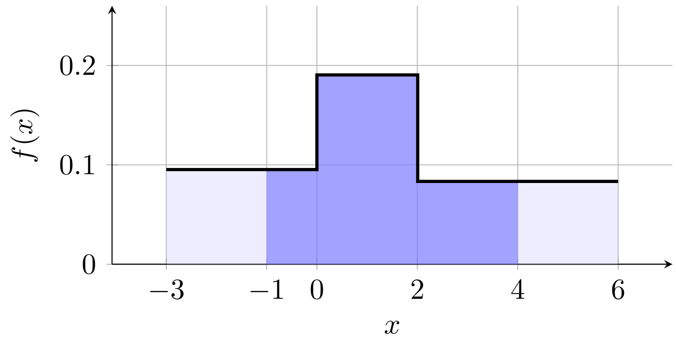
Use total area:

$$3k+2(2k)+4\left(\frac{1}{12}\right)=1\Rightarrow 7k+\frac{1}{3}=1\Rightarrow 7k=\frac{2}{3}\Rightarrow k=\frac{2}{21}.$$

$$P(-1\le X\le 4)=1\left(\frac{2}{21}\right)+2\left(\frac{4}{21}\right)+2\left(\frac{1}{12}\right)=\frac{2}{21}+\frac{8}{21}+\frac{1}{6}=\frac{13}{21}+\frac{1}{6}=\frac{11}{14}.$$

$$P(X<0)=3\left(\frac{2}{21}\right)=\frac{2}{7},\qquad P(X\ge 2)=4\left(\frac{1}{12}\right)=\frac{1}{3}.$$

So $P(X\ge 2)$ is larger.



Interpretation. A wider interval with a moderate height can beat a narrower interval with a higher height.

17 Symbolic Endpoint (Conceptual Effect of Width)

Problem. Let $m > 0$ and define

$$f(x) = \begin{cases} \dfrac{1}{3m}, & 0 \le x < m, \\ \dfrac{2}{3m}, & m \le x \le 2m, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks. Find:

(a) verify that f is a valid PDF,

(b) $P\left(X < \dfrac{3m}{2}\right)$,

(c) explain what happens to this probability if m is doubled.

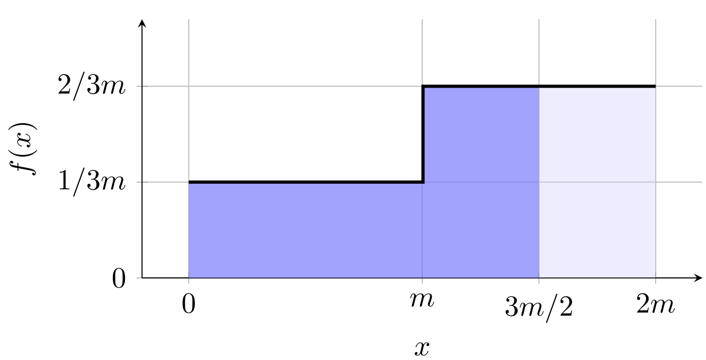
C2

$$m\left(\frac{1}{3m}\right) + m\left(\frac{2}{3m}\right) = \frac{1}{3} + \frac{2}{3} = 1,$$

so f is valid and nonnegative.

$$P\left(X < \frac{3m}{2}\right) = m\left(\frac{1}{3m}\right) + \frac{m}{2}\left(\frac{2}{3m}\right) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

If m is doubled, both widths scale by the same factor and densities scale down accordingly, so this probability stays $\frac{2}{3}$.



Interpretation. Rescaling the horizontal axis can leave certain relative probabilities unchanged.

18 Infer Missing Endpoint from a Probability Condition

Problem. Let $b > 2$ and let

$$f(x) = \begin{cases} c, & 0 \le x < 2, \\ \dfrac{c}{2}, & 2 \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Suppose $P(X > 2) = \dfrac{1}{3}$.

Questions/Tasks. Find:

(a) c,
(b) b,
(c) $P(1 \le X \le 3)$.

C2

Given

$$P(X > 2) = (b-2)\left(\frac{c}{2}\right) = \frac{1}{3}. \tag{1}$$

Total area is 1:

$$2c + (b-2)\left(\frac{c}{2}\right) = 1.$$

Using (1):

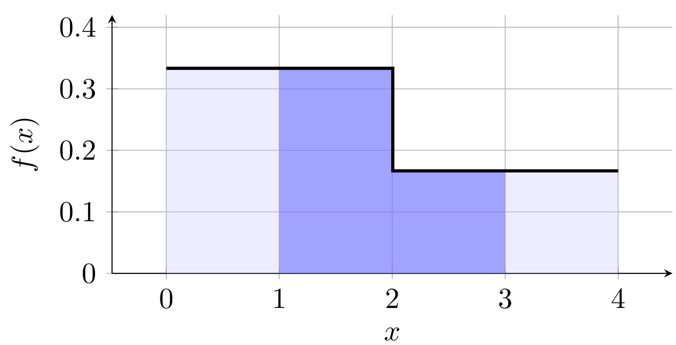
$$2c + \frac{1}{3} = 1 \Rightarrow 2c = \frac{2}{3} \Rightarrow c = \frac{1}{3}.$$

Now substitute into (1):

$$(b-2)\left(\frac{1}{6}\right) = \frac{1}{3} \Rightarrow b - 2 = 2 \Rightarrow b = 4.$$

Finally,

$$P(1 \le X \le 3) = 1\left(\frac{1}{3}\right) + 1\left(\frac{1}{6}\right) = \frac{1}{2}.$$



Interpretation. A probability condition plus total-area condition can determine unknown parameters.

19 Rich Piecewise Uniform Design Problem

Problem. Let X have density

$$f(x) = \begin{cases} k, & 0 \le x < 1, \\ 2k, & 1 \le x < 4, \\ m, & 4 \le x < 7, \\ k, & 7 \le x \le 10, \\ 0, & \text{otherwise}, \end{cases}$$

where $k, m > 0$. Assume

$$P(4 \le X < 7) = P(0 \le X < 1) + P(7 \le X \le 10).$$

Questions/Tasks. Find:

(a) k and m,

(b) $P(2 \le X \le 8)$,

(c) compare $P(1 \le X < 4)$ and $P(4 \le X \le 10)$,

(d) explain where the distribution places most of its mass and why.

C2

From the condition:

$$3m = 1 \cdot k + 3 \cdot k = 4k \Rightarrow m = \frac{4k}{3}. \tag{1}$$

Total area:

$$1(k) + 3(2k) + 3(m) + 3(k) = 1 \Rightarrow 10k + 3m = 1.$$

Using (1):

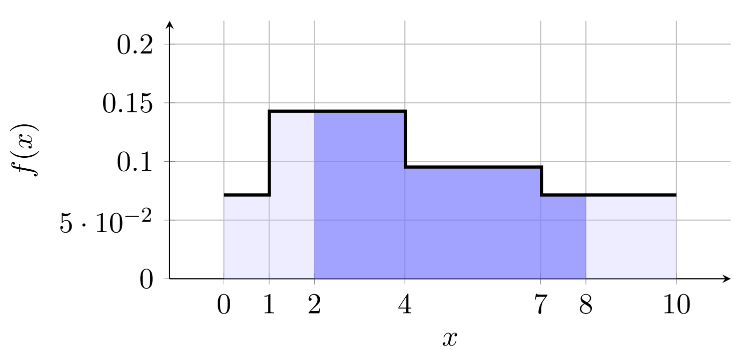
$$10k + 4k = 1 \Rightarrow 14k = 1 \Rightarrow k = \frac{1}{14}, \qquad m = \frac{4}{3} \cdot \frac{1}{14} = \frac{2}{21}.$$

Now

$$P(2 \le X \le 8) = 2\left(\frac{1}{7}\right) + 3\left(\frac{2}{21}\right) + 1\left(\frac{1}{14}\right) = \frac{2}{7} + \frac{2}{7} + \frac{1}{14} = \frac{9}{14}.$$

$$P(1 \le X < 4) = 3\left(\frac{1}{7}\right) = \frac{3}{7}, \qquad P(4 \le X \le 10) = 3\left(\frac{2}{21}\right) + 3\left(\frac{1}{14}\right) = \frac{4}{7}.$$

So $P(4 \le X \le 10)$ is larger.



Interpretation. The right side $[4, 10]$ gets more mass because it combines substantial width with moderate heights across two intervals.